PHISHGUARD//NEXUS

Explainable AI Phishing & Scam Detection Platform

AICTE – BharatCares – IBM SkillsBuild Machine Learning and Applied AI Internship 2026

Presenter & Student: Sahil Rai
Internship ID: IBMUEML002432
Academic Institution: Lovely Professional University (LPU), Punjab
Live Cloud Platform: https://phishguard-nexus.vercel.app/ (Shortlink: tinyurl.com/pg-nexus)
Source Code Repository: https://github.com/rootbysahil/phishguard-nexus

The Crisis: Zero-Day Phishing & Black-Box AI

Context & United Nations Sustainable Development Goals

Critical Industry Challenges:

- **4-Hour Attack Lifespan:** 70%+ of phishing links mutate before blacklists (Google Safe Browsing) update (12-48h lag).
- **Black-Box AI Models:** Neural nets output raw risk scores without explaining WHY, leading to user mistrust.
- **Victim IP Leakage:** Conventional scanners execute live DNS queries, leaking victim IP addresses to attackers.
- **Absence of Localized Recourse:** Non-technical Indian citizens lack plain-language advice or direct 1930 escalation.

UN Sustainable Development Goals Alignment:

- **SDG 9 (Target 9.c):** Equitable, zero-latency digital security infrastructure for all citizens & MSMEs.
- **SDG 16 (Target 16.4):** Combats financial fraud, extortion, and illicit money diversion.

System Architecture: Privacy-First & Zero-Latency

24 Pure-Python Features with Zero External Network Lookups

- **Zero-Network Feature Extraction:** 24 lexical, syntactic, and structural signals computed in <15ms with 0 DNS/HTTP calls.
- **Lexical Signals:** Shannon entropy, character run-lengths, digit & symbol ratios, URL path depth.
- **Deception Detection:** Punycode spoofing, Cyrillic homoglyph detection, brand keywords in subdomains, TLD reputation.
- **Calibrated ML Champion:** Logistic Regression with Platt scaling achieving ROC-AUC 1.000 and F2 0.9989.
- **Data Leakage Prevention:** Strict GroupShuffleSplit on registered domains eliminates test-train overlap.
- **Asynchronous SSE Engine:** Real-time pipeline milestone telemetry with EventSource streaming.

The Lean Canvas: Business & Impact Framework

Masterclass 5 Strategic Matrix

- **Unique Value Proposition:** Zero-latency forensic defense explaining WHY a target is dangerous and WHAT TO DO in regional languages.
- **Unfair Advantage:** Group-shuffled calibrated ML with zero domain leakage + native Indian Cybercrime 1930 helpline routing.
- **Customer Segments:** Indian UPI/mobile banking users, non-technical elderly citizens, and MSMEs lacking dedicated SOCs.
- **Channels:** Responsive Vercel cloud app (tinyurl.com/pg-nexus), mobile QR PWA, and AICTE/IBM showcase.
- **Revenue & Sustainability:** Free citizen tier, B2B fintech pay-per-scan API, and institutional forensic compliance audits.
- **Key Metrics:** ROC-AUC 1.000, F2 0.9989, <15ms latency, citizen scans resolved, PDF forensic audits exported.

Machine Learning Rigor: Calibrated Champion

Evaluating ROC-AUC, F2 Score, and Calibration

• **Why Logistic Regression Champion?** Uncalibrated deep neural networks are black boxes with high latency. Calibrated Logistic Regression provides transparent coefficients, zero-latency inference, and perfect calibration.
• **ROC-AUC Score: 1.0000** (Perfect discrimination across unseen domain families)
• **F2 Score (Recall-Prioritized): 0.9989** (95% CI: [0.997, 1.000]) — Penalizes missed phishing links 5x over false alarms.
• **Recall: 1.0000** (Zero false negatives on evaluation test split)
• **Precision: 0.9989** (Virtually zero benign URL disruption)
• **Domain Isolation:** GroupShuffleSplit on registered domains ensures zero synthetic inflation.

Explainable AI (XAI): Directional Attribution

Moving Beyond Black-Box Predictions to Transparent Forensics

- **The XAI Requirement:** Under IBM AI Ethics and NIST AI RMF, models must explain individual decisions to users.
- **LinearExplainer Integration:** Directional feature attribution maps exactly how each lexical signal shifts the probability.
- **Transparent Indicators:**
  - Brand keyword in subdomain: +42% Phishing contribution
  - Homoglyph / Cyrillic spoofing detected: +38% Phishing contribution
  - High-abuse TLD (.tk, .xyz): +28% Phishing contribution
  - Valid HTTPS encryption: -15% Safe indicator
- **Visual HUD Representation:** Color-coded SHAP attribution bars displayed in real time on the frontend dashboard.

IBM watsonx.ai & Granite LLM Integration

Multilingual Plain-Language Cybersecurity Guidance

- **The Advisory Bridge:** Converts mathematical SHAP weights into human-actionable cybersecurity advisories.
- **Why-Flagged Analysis:** Explains the precise technical deception in everyday language without security jargon.
- **What-To-Do Action Plan:** Provides clear, immediate instructions: *Do not enter credentials, disconnect session, report to helpline.*
- **Vernacular Language Support:** On-demand translation into **Hindi (■■■■■)** and **Punjabi (■■■■■■)** for nationwide inclusivity.
- **Reliable Fallback Architecture:** Template-based forensic fallback ensures 100% operational uptime if LLM quota is reached.

Forensic PDF Reports & National Cyber Defense

Cryptographic SHA-256 Audits & Indian Cybercrime Portal

- **Instant PDF Forensic Export:** One-click ReportLab PDF generation for enterprise incident response and law enforcement.
- **Cryptographic Proof:** Every report is stamped with an immutable SHA-256 hash of the submitted content.
- **Complete Evidence Dossier:** Includes URL anatomy, risk probability dial, SHAP breakdown, and red-flag catalog.
- **National Cybercrime Portal Routing:** Direct one-click link to **cybercrime.gov.in**.
- **National Helpline 1930 Badge:** Prominently surfaced on mobile and desktop for immediate victim intervention.

Live Production Deployment & Verification

Vercel Serverless Architecture & Global Availability

- **Live Cloud Platform:** https://phishguard-nexus.vercel.app/
- **Mobile Shortlink:** tinyurl.com/pg-nexus
- **Automated Test Suite:** 8/8 Unit & Integration tests passing 100% (pytest).
- **Zero Local Dependency:** Entire backend, ML pipeline, static frontend, and SQLite database run serverless in the cloud.
- **Mobile PWA Support:** Responsive viewport layout with 'Add to Home Screen' capability for smartphone testing.
- **Interactive API Documentation:** Swagger UI available live at /docs.

Conclusion & Phase 2 Innovation Roadmap

Delivering Future-Ready Applied AI in Cybersecurity

Internship Achievements:

• Built an end-to-end, production-grade XAI cybersecurity platform fulfilling all 6 IBM SkillsBuild masterclasses.

• Solved the dual challenge of zero-day detection and model explainability without privacy compromise.

Phase 2 Future Scope:

• **Quishing (QR Code Phishing) Defense:** Direct mobile camera decoding and malicious payload interception.

• **Counterfactual 'What-If' Simulator:** Interactive dials allowing users to simulate feature changes and see live probability shifts.

• **Crowdsourced Threat Radar:** Federated threat intelligence exchange for Indian educational institutes and MSMEs.

Thank You! Ready for Evaluator Q&A.;